Mitchell Eickhoff

Systems & Security Engineer | Embedded Firmware | Reverse Engineering

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SUMMARY

Systems-focused engineer with hands-on experience in embedded firmware security, reverse engineering, and low-level systems architecture; from voltage-glitch attacks in capture-the-flag competitions to a compiler backend contribution merged into a production C toolchain.

EDUCATION

Ohio State University — B.S. in Computer Science & Engineering

May 2028 | 3.66 GPA

TECHNICAL SKILLS

Security/Embedded: Firmware development, reverse engineering, dynamic hooking, web exploitation, secure coding, ARM & Z80 assembly, RTOS/Microcontrollers, Firmware Development

Languages: C, Z80 Assembly, ARM Assembly, Rust, Python, JavaScript/TypeScript, Java, SQL

Systems: Linux, shell scripting, memory paging, compiler backends, low-level hardware interop

Web/AI: Next.js, FastAPI, WebRTC, PyTorch, Whisper, LLM integration

Tools: Git, z88dk, FFmpeg, Tampermonkey, PostgreSQL, SQLite

PROJECTS

Ti-Tale *Systems Architecture, Z80 Assembly, Reverse Engineering*

- Designing a custom Z80 game engine for TI-83 Plus hardware, with hand-written assembly for interrupt handling, RAM paging, cooperative multitasking, and flash memory management
- Reverse-engineered undocumented OS and hardware behavior to safely interoperate with the calculator's operating system

github.com/PsychedelicPalimpsest/TI-tale

Z88DK Compiler Contribution *C, Z80 Assembly, Compilers, Memory Paging, GMP*

- Engineered multi-page Flash Application (.8xk) compilation support for TI-83+/TI-84+ targets in the Z88DK C compiler toolchain
- Extended the compiler backend and Z80 runtime to auto-generate cross-page banked calls, and built a packaging pipeline for hardware branch tables and GMP-based cryptographic signing required by TI-OS

github.com/z88dk/z88dk/pull/2433

LibbyRip *Web Exploitation, Reverse Engineering, Dynamic Hooking*

- Built a resilient tool for extracting data from the Libby platform by hooking global JavaScript functions (JSON.parse, Function.prototype.bind) to passively capture application state
- Grew to 1,000+ active users and 300+ GitHub stars; added on-the-fly media conversion via a WebAssembly build of FFmpeg

github.com/PsychedelicPalimpsest/LibbyRip

(Find more on my portfolio and github)

EXPERIENCE

eCTF Club Member The Ohio State University *2025 – Present*

- Compete in MITRE's Embedded Capture the Flag competition, writing secure firmware in C and exploiting competing teams' code
- Apply working knowledge of ARM assembly and hardware attack vectors, including voltage-glitch attacks

Open Source Developer *2015 – Present*

- Build and maintain dozens of open-source projects across C, Rust, Python, JavaScript, Java, and Z80 Assembly; contribute to third-party toolchains at the compiler and runtime level (see Z88DK project above)

MIS Intern Ohio Public Defender's Office *Oct 2023 – May 2024*

- Supported development of OPD Online, a statewide case management system used across all 88 Ohio counties
- Gathered requirements from attorneys and staff and coordinated a staged statewide rollout plan with outsourced contractors

On-Campus Instructor iD Tech *Summer 2026*

- Taught Blender 3D modeling fundamentals to students ages 12–17, building technical communication skills

CSE 2421 Grader OSU *Fall 2026*

- Graded student programming projects in C and X86 assembly, providing clear, actionable feedback on correctness, style, and low-level systems concepts
- Held regular office hours to help students debug assignments, strengthen their understanding of course material, and improve problem-solving skills